

Ex. No 5 Use Autopsy to create a case and import evidence

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link:

https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

Download files: 4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02

1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
- Click on New Case.
- Enter the case name and location where the case data will be stored.
- Fill in the details like the case number, examiner's name, etc., and click Next.

The screenshot shows the 'New Case Information' window in the Autopsy application. On the left, a 'Steps' sidebar lists '1. Case Information' and '2. Optional Information'. The main window is titled 'Case Information' and contains the following fields and options:

- Case Name:** A text box containing 'siddartha'.
- Base Directory:** A text box containing 'C:\Users\siddu\OneDrive\Desktop\iii\' with a 'Browse' button to its right.
- Case Type:** Two radio buttons, 'Single-User' (which is selected) and 'Multi-User'.
- Case data will be stored in the following directory:** A text box containing 'C:\Users\siddu\OneDrive\Desktop\iii\siddu'.

At the bottom of the window, there are five buttons: '< Back', 'Next >' (highlighted in blue), 'Finish', 'Cancel', and 'Help'.

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case Number: 99240041228

Examiner Name: siddartha

Phone:

Email:

Notes:

Organization: Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

3. Adding a Data Source

- **Choose the Type of Data Source:**
- After creating a case, you'll be prompted to add a data source.
- You can add disk images, directories, logical files, or local disks.**Select the Data Source:**
- Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
- <https://drive.google.com/drive/u/1/folders/1lSFY7Tqn2L7AjQGhq8yJ8kixcxTU-v>

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. **Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Local files and folders

C:\Users\siddu\Downloads\4Dell Latitude CPI.E01

C:\Users\siddu\Downloads\4Dell Latitude CPI.E02

Add Delete Clear

Logical File Set Display Name: Default Change

Timestamps To Include:

☐ Modified Time - Often not changed when a file is copied

☐ Creation Time - Often changed when a file is copied

☐ Access Time - Can be changed when the file is opened

NOTE: Time stamps may have changed when the files were copied to the current location.

< Back Next > Finish Cancel Help

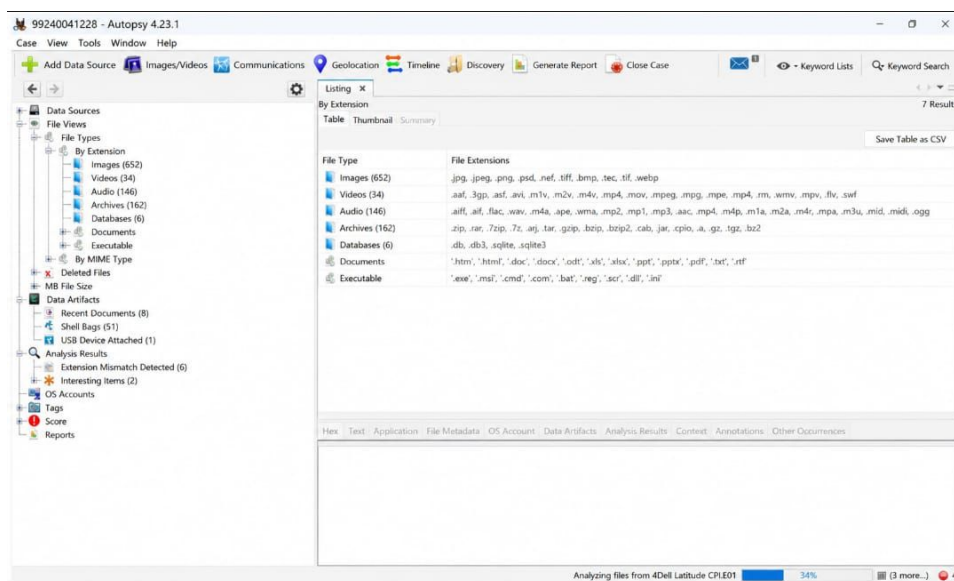
- **Configure Ingest Modules:**
- Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
- You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.

4. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
- Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- **Use the Tree Viewer:**
- On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**
- You can perform specific keyword searches using the Keyword Search module.
- Use pre-configured lists or enter custom keywords.
- **File Analysis:**



- Navigate through files and folders under the File Types or File System section.
- Open, view, or export files for further examination.
- **Timeline Analysis:**
- Use the Timeline module to visualize events based on timestamps.
- This can help track user activity over time.
- **Hash Analysis:**
- Compare file hashes with known databases to identify known good or bad files.

6. Reporting

- **Generate a Report:**
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.



7. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.s

report on:

ts
Results

types...

 Result Type Selection ✕

Select which result types you would like to report on:

☒ Accounts

☐ Data Source Usage

☒ E-Mail Messages

☐ Encryption Suspected

☐ Extension Mismatch Detected

☐ Installed Programs

☐ Interesting Items

☐ Operating System Information

☐ Recent Documents

Select All

Deselect All

OK